

AI Usage Note

Tools Used

I used **ChatGPT** as an AI assistant during this project.

How I Used AI

I mainly used ChatGPT to:

- Understand the requirements of the challenge.
- Break the research into smaller steps.
- Help write and debug Python code.
- Understand concepts such as confidence intervals, hypothesis testing, bootstrap, baseline comparison and look-ahead bias.
- Help structure the Research Note and README.

My Own Decisions

I made the main research decisions myself, including:

- Choosing the **-3% event threshold**.
- Choosing **1D, 3D and 5D** holding periods.
- Using the **next trading day's Open** as the entry price.
- Choosing the baseline and robustness checks.
- Reviewing and interpreting the final results.

I ran the code on the NIFTY dataset and checked the outputs before using them in the project.

Suggestions I Changed or Disagreed With

During the project, I identified that an initial baseline approach could give incorrect forward returns if event rows were removed before using `shift()`. I changed the approach so that forward returns were calculated on the complete trading-day timeline before separating event and non-event observations.

I also learned that positive returns or a high win rate alone do not prove that an event effect is statistically meaningful.

What I Learned

Using AI helped me understand the research process and statistical concepts more clearly. However, I treated AI suggestions as assistance rather than as final answers and checked the code and results myself.

Final responsibility for the methodology, analysis and conclusions remains with me.